



Sanskar Ranjan

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B.Tech - Mathematics and Computing

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Guwahati	7.89 (Current)	2024-Present
Senior Secondary	CBSE Board	97.8%	2024
Secondary	CBSE Board	97.6%	2022

PROJECTS

- VELORA: Fleet Routing & Optimization Engine** Jan. 2026 – Mar. 2026
Gold Medal, Kriti 2026, Inter Hostel Technical Competition, IIT Guwahati bit.ly/opti-Kriti bit.ly/opti-web
 - Architected a **Fleet Optimization System** that automates multi-vehicle dispatching, using a **dual-objective function** to minimize both financial costs and passenger transit times while enforcing spatio-temporal constraints.
 - Built a **multithreaded C++ backend (Crow REST)** with **OSRM/Haversine** distance-matrix pipeline, concurrently executing **routing metaheuristics** (VNS, ALNS, Branch & Cut) in isolated sandboxes to get **80% fleet cost savings**.
 - Developed a **Next.js React frontend** to translate complex heterogeneous Dial-A-Ride constraints into interactive geospatial visualizations, seamlessly deploying the complete full-stack architecture on **GCP and Vercel**.
- MediRAG: Agentic Mental Health Assistant** Mar. 2026 - Apr. 2026
Group project, 4i Labs, IIT Guwahati bit.ly/multimodal-4i
 - Developed an AI-powered clinical mental health assistant that tracks symptom data and delivers medically verified coping strategies using a **Hybrid RAG pipeline (PyMuPDF, BM25, ChromaDB)** with **Cross-Encoder re-ranking**.
 - Architected a stateful **FastAPI** and **LangGraph** backend to manage conversational memory, utilizing **LangChain** and **Pydantic** tools to extract unstructured patient chat logs into structured fields stored in **Neon PostgreSQL**.
 - Implemented a custom "LLM-as-a-judge" evaluation suite to benchmark the RAG pipeline against a synthetic clinical dataset, achieving mathematically-verified metrics of **0.88 Faithfulness** and **0.86 Answer Relevancy**.
- CVM++: Stack-Based Virtual Machine & Custom Compiler** Apr. 2026 - May. 2026
Group Project, Coding Club, IIT Guwahati bit.ly/cvm-plus
 - Architected a custom scripting language infrastructure from scratch in C++, incorporating a **tokenizing lexer**, a top-down recursive-descent **parser**, a **bytecode compiler**, and a **stack-based VM** for script execution.
 - Built an **Abstract Syntax Tree** using a recursive-descent parser, leveraging C++ **RAII principles** for strict node ownership, ensuring memory-safe runtime evaluation of operators without relying on garbage collection.
 - Programmed a bytecode compiler that converts AST nodes into executable instructions using **jump backpatching**, powering a runtime engine engineered to track global variables and opcodes, and to maintain stack frames.

ACHIEVEMENTS

- Competitive Programming**, Max Rating of **1530** on Codeforces (**Specialist**) | Handle: *ranjan.sanskar*
 - Codeforces Round 1090 (Div. 4): Secured **Global Rank 543** among 20,000+ participants
 - Codeforces Round 1093 (Div. 2): Secured **Global Rank 1365** among 20,000+ participants
- Kriti - Inter Hostel Tech Competition**, Won **Gold Medal** in Optimization Challenge 2026
- Indian Olympiad Qualifier in Mathematics**, Qualified for **INMO** and awarded a **Merit Certificate**. 2022
- NSEP (National Standard Examination in Physics)**, Ranked in **National Top 1%** out of 49k+ candidates. 2024
- NSEA**, Qualified for INAO 2024, ranking among **top 258** out of 16,000+ candidates nationally. 2024
- NSEC**, Qualified for INChO 2024, ranking among **top 531** out of 44,000+ candidates nationally. 2024
- NSEJS Merit Certificate Holder**, Ranked in **Top 300** among 10,000+ national participants. 2021

TECHNICAL SKILLS

- Programming:** C/C++, Python
 - Web Technologies:** HTML, CSS, JavaScript*, ReactJS*
 - Machine Learning Frameworks:** PyTorch, NumPy, LangGraph, TensorFlow*
 - Operating Systems:** macOS, Windows, Linux*
 - Miscellaneous:** MySQL, Gem5, Git, $\text{\LaTeX}$ *, Verilog
- * Elementary proficiency

KEY COURSES AND CERTIFICATIONS

- Computer Science:** Design and Analysis of Algorithms, Database Management Systems (+Lab), Computer Organization and Architecture, Digital Design, Data Structures (+Lab), Introduction to Computing (+Lab)
- Mathematics:** Probability Theory and Random Processes, Discrete Mathematics, Linear Algebra, Partial Differential Equations, Elementary Number Theory and Algebra, Financial Engineering
- Certifications:** Deep Learning Specialization, Machine Learning Specialization (DeepLearning.AI / Andrew Ng), BNY Spectrum Program, MachineLearning.AI(2025), Summer Analytics(2025)